

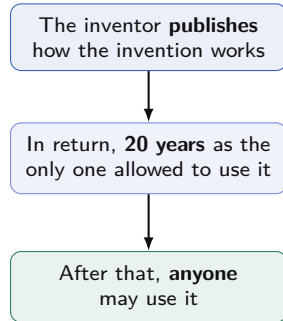
Is “Intellectual Property” Really Property?

The word **property** makes an idea sound like a car or a phone: something you own, and something someone else can take from you.

A patent does not work like that. It is a **trade**. The inventor publishes how the invention works, and in return gets **20 years** as the only one allowed to use it. After that, it is free for anyone.

And the **idea** by itself is never covered. A theorem, a plot, or a technique cannot be patented or copyrighted by anyone.

A patent is a temporary deal rather than permanent ownership.



The patent bargain: disclosure now, exclusivity for a while, then it belongs to everyone.

Why Taking Things Is Wrong: What if Everyone Did It?

The generalization test, from Lecture 11

An action is wrong if the reason for doing it stops working once everyone does it.

Stealing fails that test:

- The thief takes someone's property because he wants **possession and use** of it.
- But if no one respects property rights, the thief **won't have** possession and use either.
- In effect, there would be **no property** at all.

This works because physical goods are exclusive: if I have it, you do not.



A private property sign. Possession by one person rules out possession by another.

Copying Is Not Taking

But IP is **not** like other property: one can use IP **without denying others the use of it**.

Physical property

One car, one owner at a time.
Taking it removes possession
and use from someone else.

Intellectual property

Millions can possess and use the
same song, text, or design at once.
Nobody loses their copy.

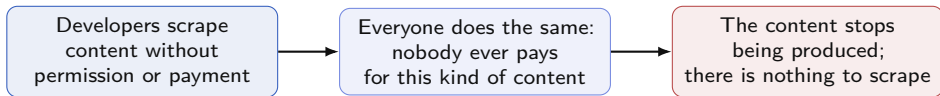
So the standard argument against theft does not, by itself, explain why copying is wrong.

Two students obtain the same \$80 textbook without paying. **A** takes a classmate's printed copy off her desk. **B** downloads a scanned PDF from a group chat, and the classmate keeps hers. Does the test condemn both, one, or neither? If B did something wrong, what argument are you using?

Scraping Fails a *Different Test*

Scraping is how these models get their training data: software automatically downloads text and images in bulk from public websites. Nobody is asked, and nobody is paid.

None of this says there is **no reason** to respect IP. A **second** generalization argument applies here, and it is the “garbage loop” pattern from last lecture: a practice that destroys what it feeds on.



On this argument it does not matter whether the output **resembles** the input, so the images we opened with are the wrong thing to be angry about.

What scraping takes is the money that pays for the reporting, which makes this a question about how journalism gets funded rather than about plagiarism.

Intellectual Property Law: The TRIPS Agreement

IP law differs in every country. The primary international treaty is the **TRIPS agreement**: Trade-Related Aspects of Intellectual Property Rights, dating from 1994 and administered by the **WTO**.

- It sets the **minimum** protection each member must provide. National law may go further, but not less far.
- The TRIPS minimums are life plus 50 years for copyright, and 20 years for a patent.
- Treaties set the standards; **national offices** grant and enforce the actual rights, such as the U.S. Patent and Trademark Office.



TRIPS is a floor, not a ceiling.

Four Kinds of Intellectual Property

Each protects something different, and they differ in how you obtain them and how long they last.

	What it protects	How you get it	How long it lasts
Copyright	A particular expression: code, text, image, melody	Automatically, on creation	Life + 50 years (TRIPS); 95 years for corporate works in the U.S.
Trademark	A mark identifying the source of goods or services	Registration, or commercial use	Indefinitely, while still used commercially
Patent	A new method, device, or material	Application and examination, in exchange for disclosure	20 years
Trade secret	Secret information giving commercial advantage	By keeping it secret	Until the secret escapes

Only copyright and patent expire on a clock. A trademark dies of disuse, a trade secret of disclosure.

Fair Use

Is it legal to download this cartoon?

Copyright is not absolute. Certain uses of a copyrighted work are not infringement at all. Section 107 of the U.S. Copyright Act:

*"The **fair use** of a copyrighted work . . . for purposes such as criticism, comment, news reporting, **teaching** (including multiple copies for classroom use), scholarship, or research, is **not** an infringement of copyright."*

Whether a use is fair turns mainly on its **purpose**: commercial, or nonprofit and educational.

The same copy can be fair use in a classroom and infringement in a product.



"I KNOW IT'S ILLEGAL TO DOWNLOAD MUSIC FROM THE INTERNET... BUT YOU SAID THE STUFF I LISTEN TO ISN'T MUSIC!"

Cartoon © 2004 Randy Glasbergen, glasbergen.com. Reproduced here for classroom teaching.

Can Software Be Copyrighted?

Generally, **yes**.

- The TRIPS agreement explicitly says yes.
- But one **cannot** copyright the underlying idea or algorithm.

**Your source code is protected as an expression.
The technique it implements is not, so someone
else may reimplement it from scratch.**

```
/*  
 * Copyright (c) 2026 Example Ltd.  
 * All rights reserved.  
 *  
 * Permission is hereby granted, free  
 * of charge, to any person obtaining  
 * a copy of this software ...  
 */  
  
int sort(int *a, int n) {  
    /* the algorithm: not protected */  
}
```

The header and the code are a literary work and are protected.

The sorting method the function implements is not.

Trademark

A **trademark** is a word, phrase, symbol, or design that **distinguishes** the maker or source of goods. A **service mark** serves the same function for a service.

- Established by **registration**, or in common law by commercial use.
- Rights can last **forever**, as long as the mark stays in **commercial use**.
- One cannot claim a mark **already registered or in use** by another company in the same product class.



Marks identify the source of a product, so a buyer knows who is answerable for it.

Two Apples

Apple Corps, the Beatles' record label, registered its mark before Apple Computer existed. Both kept the name, because records and computers are different **product classes**. That held until Apple Computer started selling music, and the dispute ran from 1978 to 2007.

Patent

A patent is designed to encourage **disclosure** of ideas in exchange for a limited period of exclusive use.

- **20 years** in TRIPS.
- In the U.S.: **20 years**, except **15 years** for a design patent.

Can patent: a **new** method, device, or material.

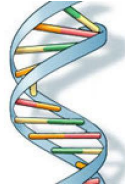


An invention already known to the public fails the novelty requirement and cannot be patented.

What Cannot Be Patented

- A pure idea, such as a theorem.
- Anything that occurs in nature.
- A “way of doing business,” even if automated by computer.
- “Look and feel,” for example of a spreadsheet.

Each exclusion keeps something available to everyone: mathematics, nature, ways of doing business, and interface conventions.



Naturally occurring sequences and the look and feel of an interface both fall outside patent protection.

Can Software or an Algorithm Be Patented?

Controversial, and it varies by country. The TRIPS agreement is **vague**. In the U.S. one can patent:

- a **software-based invention**;
- software that improves **computer functionality**;
- software that solves a problem arising only in the context of **computer technology**.

An algorithm **in principle** cannot be patented: it must be embodied in some device. **In practice** many have been, since a patent is issued not for source code but for a “series of steps,” which is an algorithm. **Bubble sort**, however, was denied a patent.

Ask a **patent lawyer**; it is a recognized specialty.

Patentable

“a faster way for a computer to do X”

Not patentable

“doing X, but on a computer”

Patentability turns on what problem the software is said to solve, not on the code itself.

Who Owns a Patent?

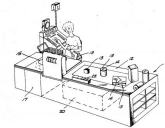
A patent is **registered** in the name of the inventor.

The **owner** may be someone else, or a company. An employer normally **owns** any idea conceived by someone working **for hire**.

Being named as the inventor and owning the invention are two different things.

United States Patent		(10) 3,663,800		
Myer et al.		(45) May 16, 1972		
(54) OFFICIAL LABEL READER AND DECODER	2,812,246	6/1960 Robinson		
(71) Inventors: John H. Myer, Woodland Hills, Robert L. Hastings, Culver, Francis P. Williams, San Jose, Calif.	2,812,247	6/1960 Robinson		
(72) Assignee: Hughes Aircraft Company, Culver City, Calif.	2,812,248	6/1960 Robinson		
(22) Filed: Jan. 31, 1971	2,812,249	6/1960 Robinson		
(31) Appl. No. 186,245	2,812,250	6/1960 Robinson		
(52) Continuation of Ser. No. 718,514, Mar. 27, 1968, abandoned.		2,812,251	6/1960 Robinson	
(51) Int. Cl.		2,812,252	6/1960 Robinson	
(52) Field of Search		2,812,253	6/1960 Robinson	
(53) References Cited		2,812,254	6/1960 Robinson	
(54) United States Patents		2,812,255	6/1960 Robinson	
2,612,994		10/1952 Woodland	2,812,256	6/1960 Robinson

13 Claims, 14 Drawing Figures



US Patent 3,663,800, the supermarket barcode scanner. Field

[72] names the three inventors; field [73] gives the assignee,

Hughes Aircraft, which owns it.

Trade Secret

A **trade secret** is a secret formula or device that provides a commercial advantage. It can be bought, sold, and licensed.

It remains intellectual property **forever**, or until the secret gets out, as with the formula for Coca-Cola.

A patent gives up secrecy in exchange for 20 years of protection. A trade secret keeps the secrecy and gets no fixed term.



The Coca-Cola formula was never patented, since a patent would have published it.

Nondisclosure Agreements

A **nondisclosure agreement** prevents disclosure of proprietary information.

- The **duration** of the NDA is usually stated (1–5 years). If not stated, it may be forever.
- “Proprietary” is defined in the NDA itself.
- Disclosure of **trade secrets** at any time is **illegal**, with or without an NDA.
- Generally the company identifies what it considers to be trade secrets.



The NDA fixes the scope and the duration. Trade secret law applies regardless.